



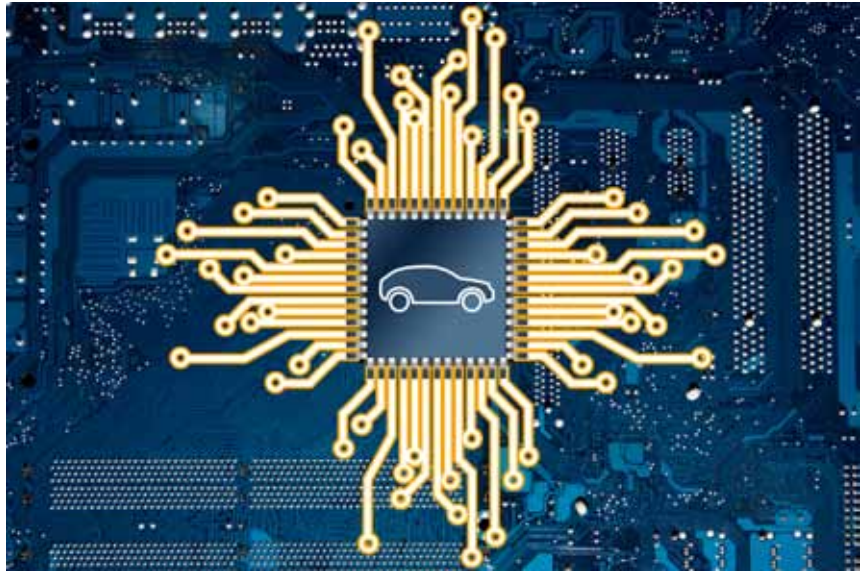
sation, higher disposable income, expanding domestic customer base, favourable demographics, supportive infrastructure, and increased foreign investment.

The landscape of the Indian automotive industry is changing rapidly with emerging business models and trends, such as connected, autonomous, shared, and electric (CASE). Technologies that support the shift toward connected, autonomous, and electric vehicles, along with customer-centric parameters, such as user experience and infotainment, are expected to boost the demand for automotive electronics in the coming years.

According to Frost & Sullivan, the Indian automotive electronics market has the potential to generate revenue of \$9.2 billion by 2025. Body control module (BCM), anti-lock braking system (ABS), in-car entertainment system (ICES), tire pressure monitoring system (TPMS), telematics, and battery management system (BMS) are some of the prominent technologies that will drive this market. About 65 to 70% of India's demand for automotive electronics is currently met through imports.

Various measures and policy incentives have been announced in recent years to boost the domestic electronics system design and manufacturing (ESDM) and strengthen the domestic automotive electronics manufacturing ecosystem. Customer expectations, regulations, and evolving mobility infrastructure are the drivers.

Evolving customer expectations. Indian automotive industry continues to evolve with increasing customer expectations from products, price sensitivity, shorter product life cycles, frequent changes in product ownership, focus on safety and entertainment features, and personalised experiences. Next-generation digital solutions will play a pivotal role in this transformation, which



will drive the automotive electronics business.

Regulatory interventions. India's vehicle safety standards have improved noticeably in recent years with next-gen technologies, such as the advanced driver assistance system (ADAS), expected to be introduced soon. Government safety standards, such as AIS 145, the adoption of telematics systems, and consumer demand for safer vehicles, will act as a catalyst for increased adoption of automotive electronics in India in the future, with estimates indicating that safety, ADAS, and infotainment will account for 60% of this demand.

Changing face of mobility infrastructure. Technologies like self-driving vehicles revolutionise the mobility sector. The development of alternative means of transportation and smart infrastructure (smart cities, parking optimisation, AI-driven traffic lights, and enabling EV-charging infrastructure) is expected to transform India's mobility infrastructure and generate the business potential for automotive electronics industry.

Next-gen technologies

Multiple next-gen technologies will be commercialised over the next

4-5 years. The passenger vehicles segment is expected to contribute two-thirds revenue of the Indian automotive electronics market. This would be driven by the increasing usage of the telematics control unit (TCU), infotainment, electronic control unit (ECU), onboard diagnostics (OBD), and anti-lock braking system (ABS). Head-up display (HUD), blind-spot detection, advanced driver assistance systems (ADAS), auto-dimming mirror, and automatic transmission features will remain mainstream for the premium segment cars.

Advancements in electronics for various vehicle components will require the installation of electronic control and communication systems. Advanced electronic systems should be directed to operate active safety features like dual airbags, electronic stability control, advanced braking systems, and tire pressure monitoring systems. Stringent regulations, such as Corporate Average Fuel Efficiency II and Bharat Stage-VI, are also significant drivers of the rising electronic content in a car. The government's focus on promoting electric mobility through various policy measures will propel the demand for automotive electronics in the coming years.